



Single-Cell RNA-seq Analysis Report

PBMC 3k Dataset - Comprehensive Analysis Pipeline

TOTAL CELLS

2,700

TOTAL GENES

13,714

HV GENES

2,000

CLUSTERS

8

MEAN GENES/CELL

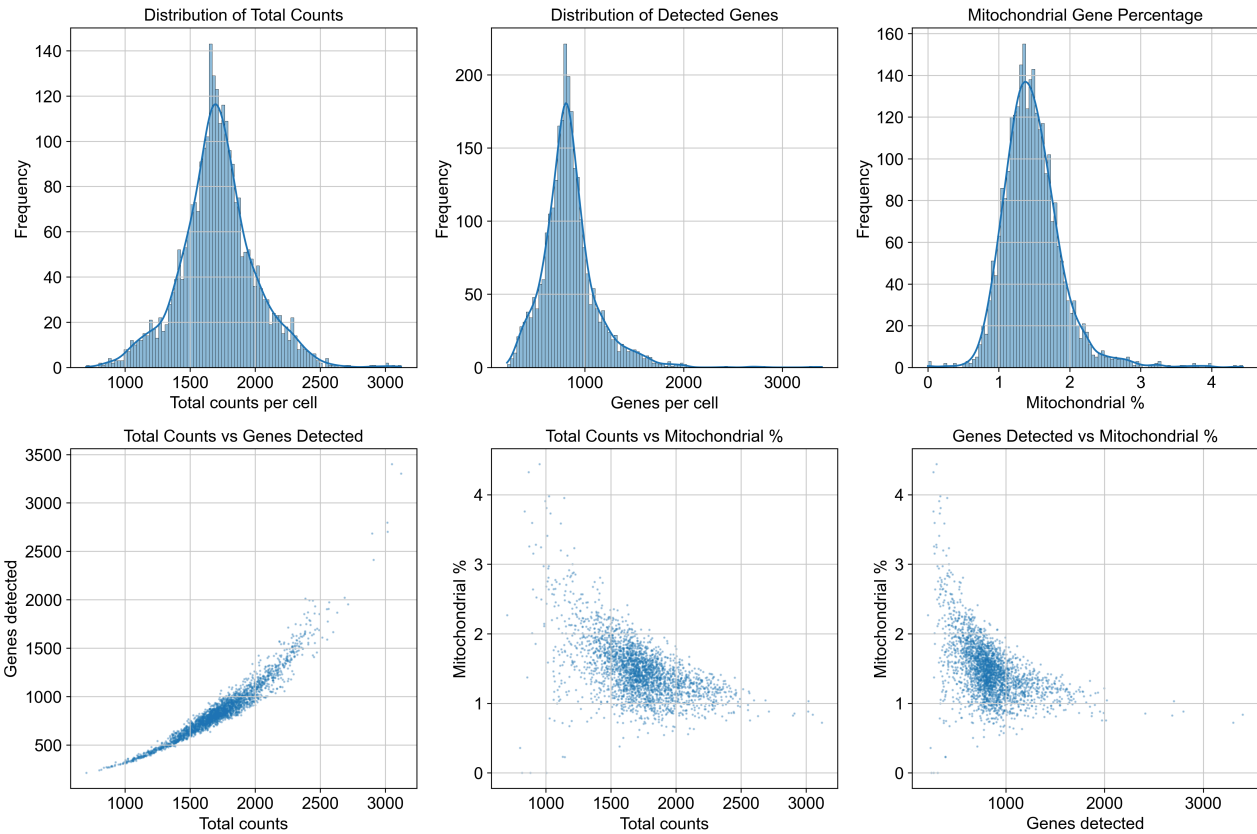
846

MEAN COUNTS/CELL

1713

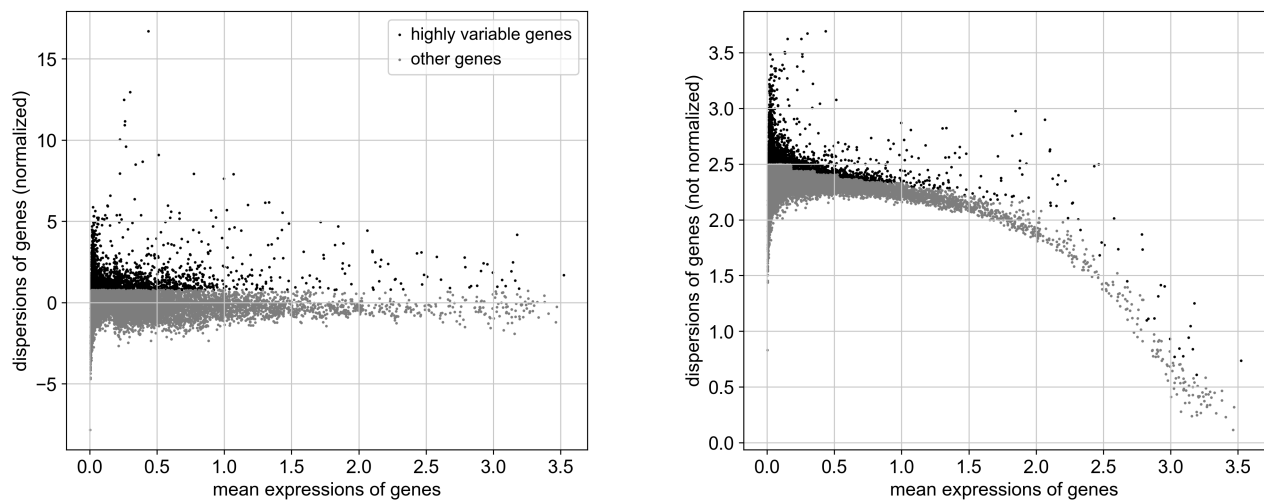
Quality Control

Distribution of QC metrics: total counts, genes detected, and mitochondrial percentage



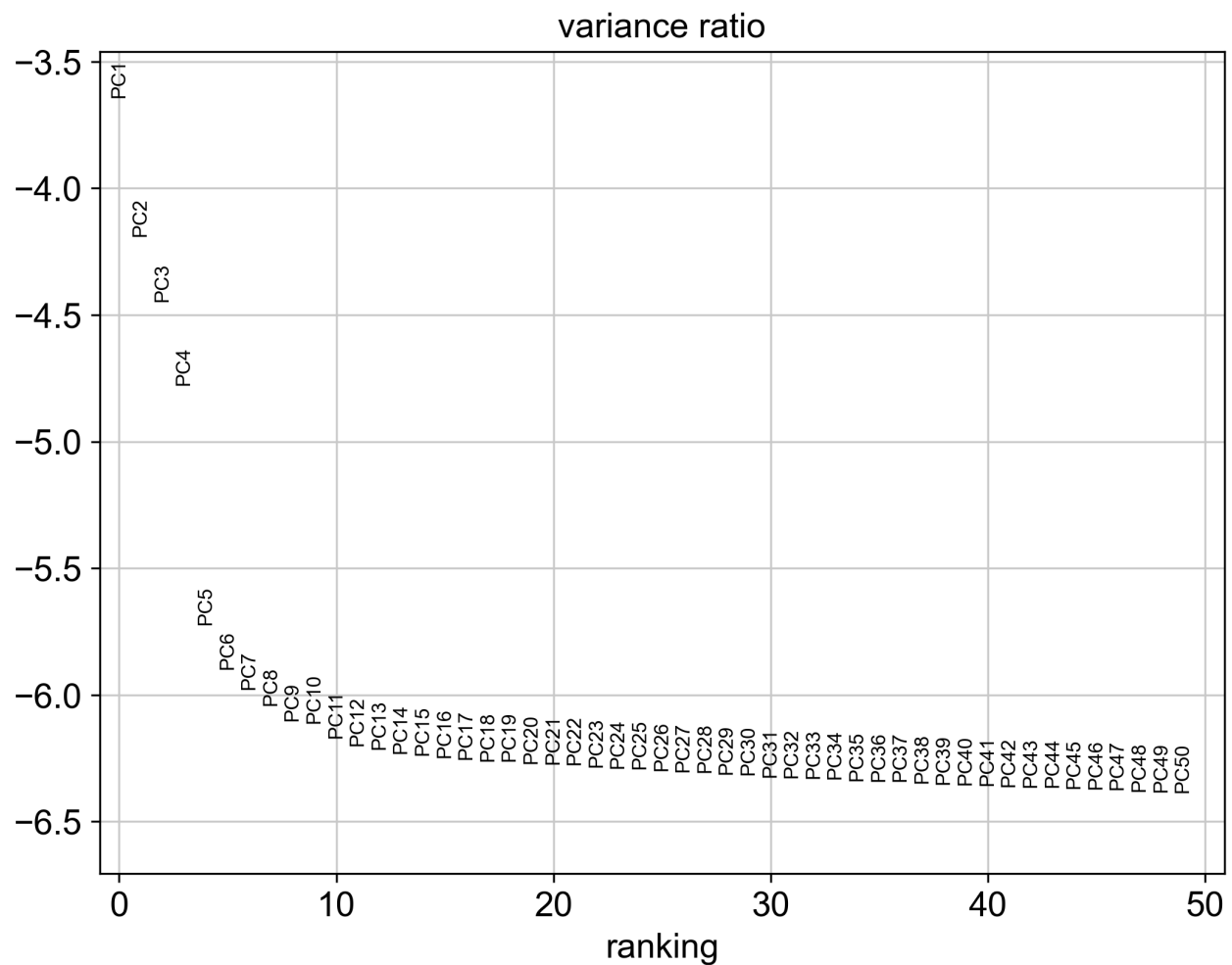
Highly Variable Genes

Identification of 2,000 highly variable genes for downstream analysis



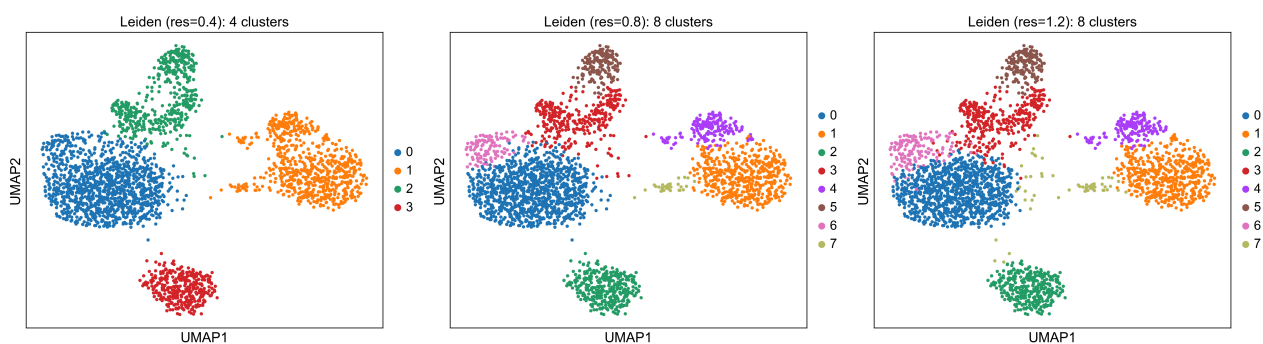
PCA Variance

Variance explained by principal components



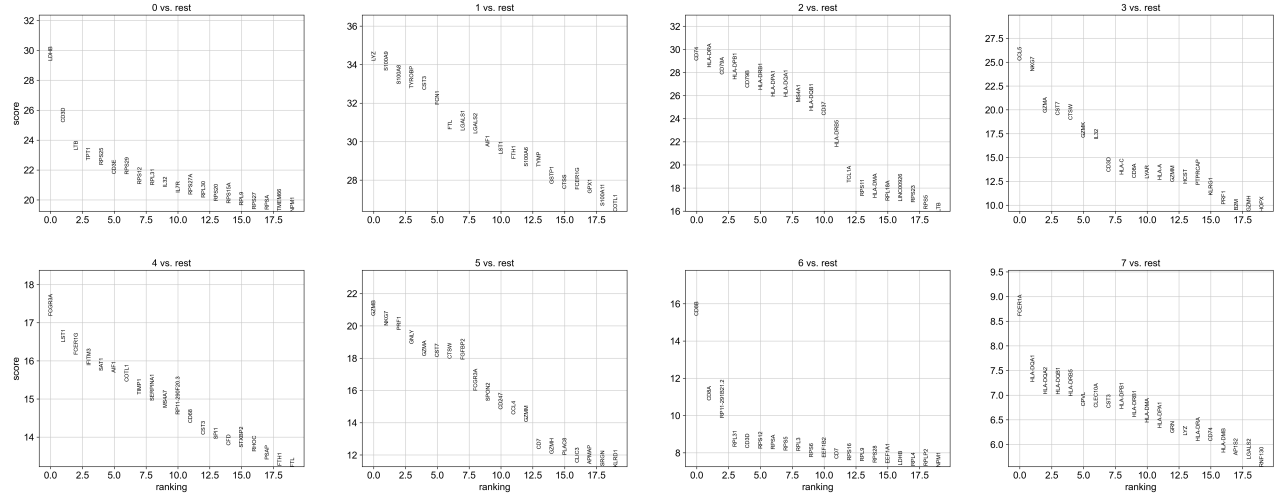
Multi-Resolution Clustering

Leiden clustering at resolutions 0.4, 0.8, and 1.2



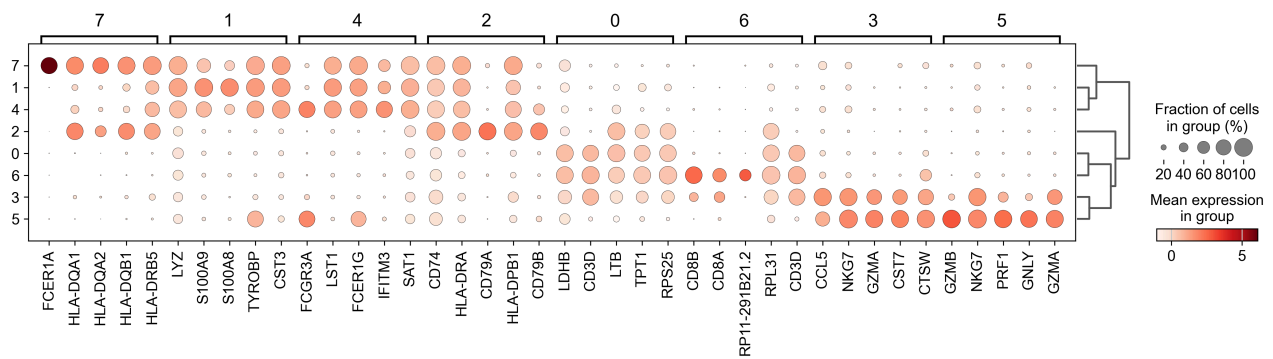
Marker Gene Ranking

Top marker genes for each cluster (Wilcoxon test)



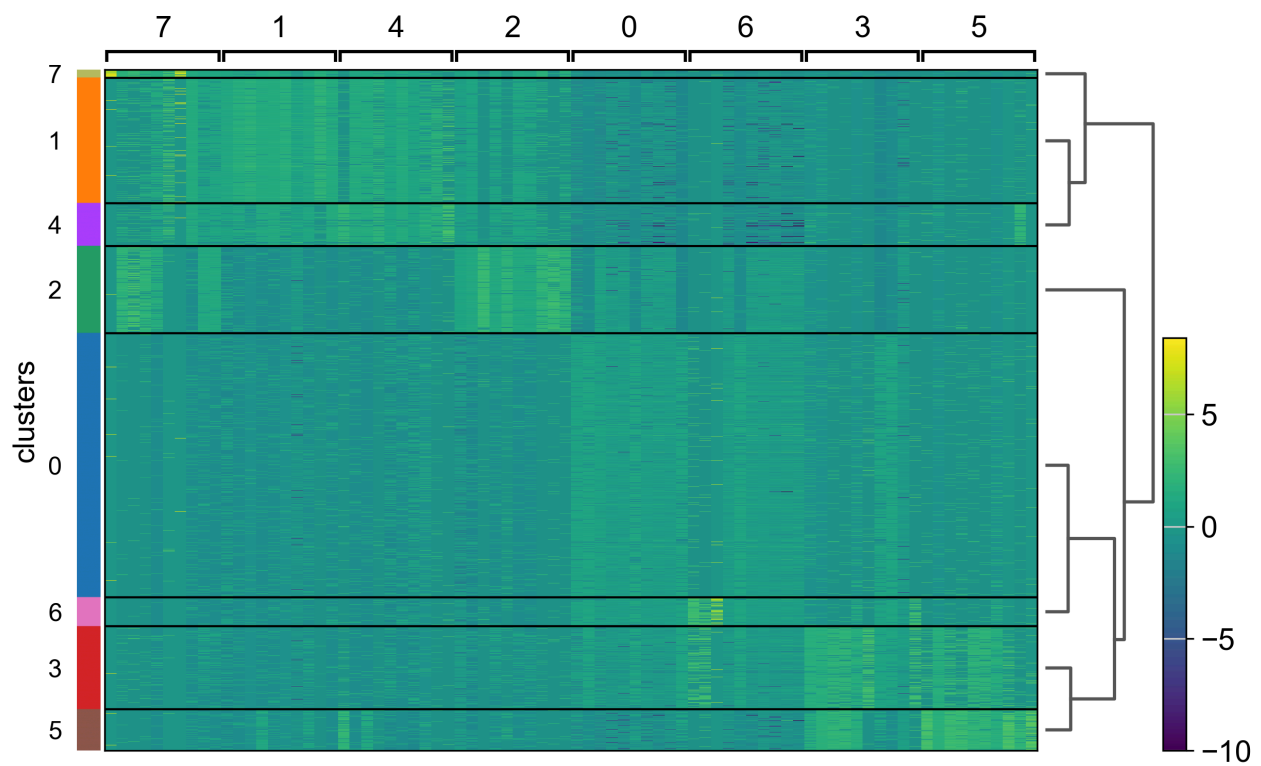
Marker Expression Dotplot

Expression of top markers across clusters



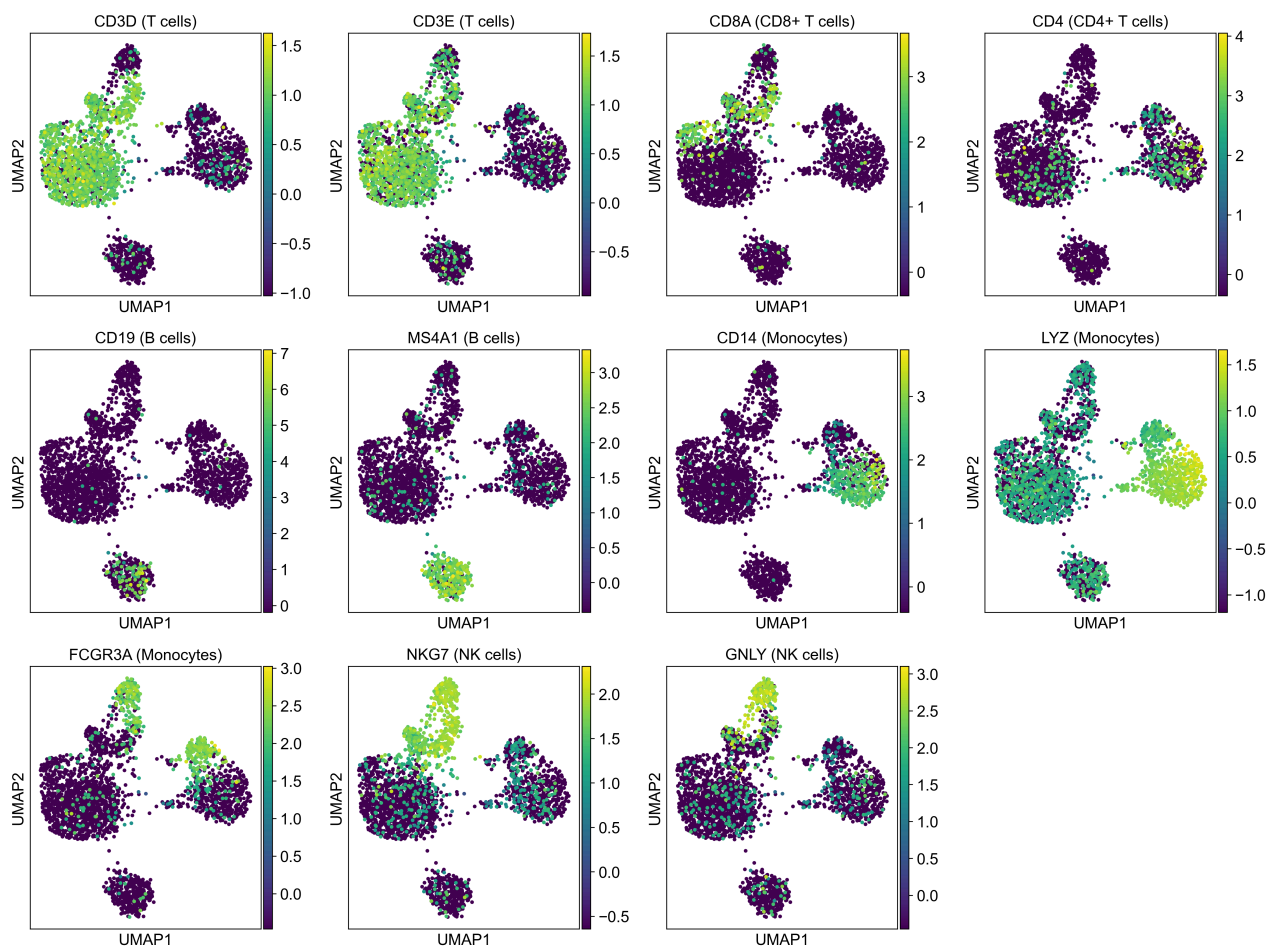
Marker Expression Heatmap

Heatmap of top marker genes per cluster



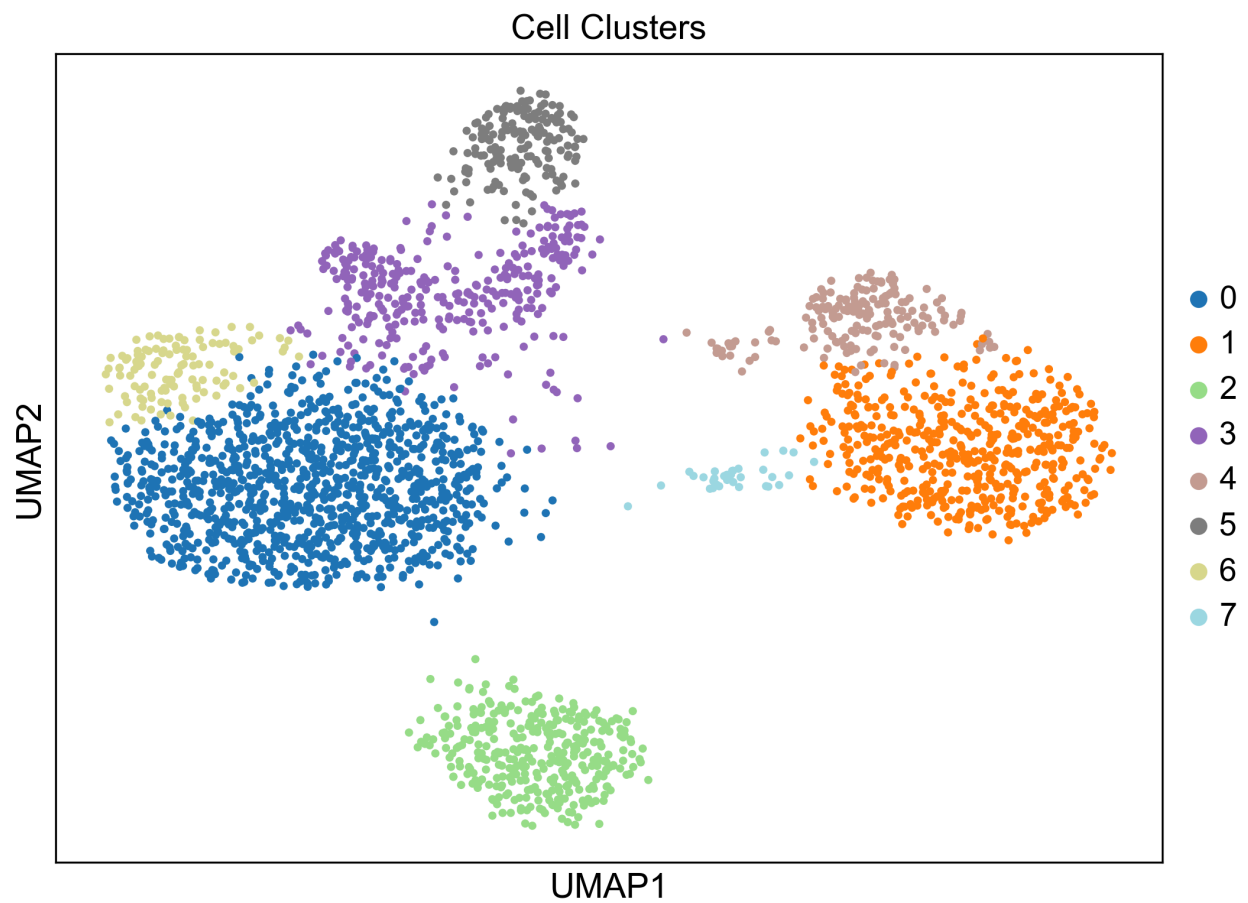
Known PBMC Markers

Expression of known cell type markers on UMAP



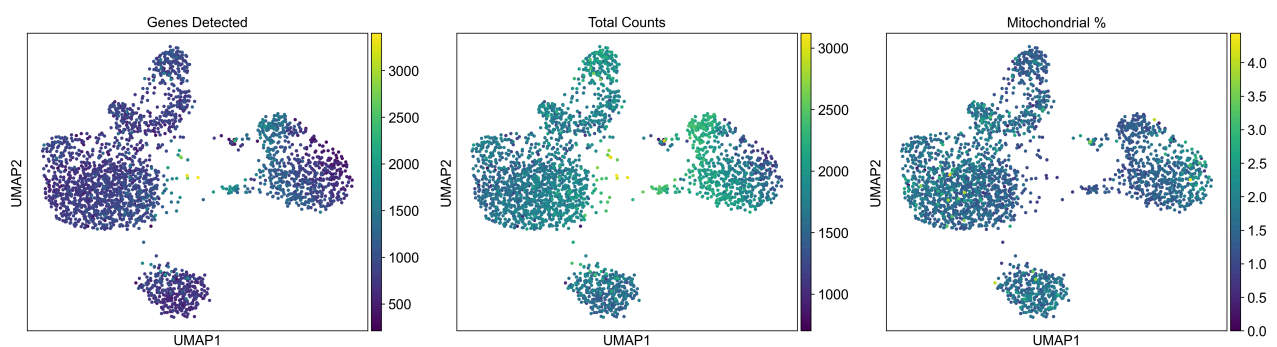
UMAP Clustering

UMAP visualization colored by cluster assignment



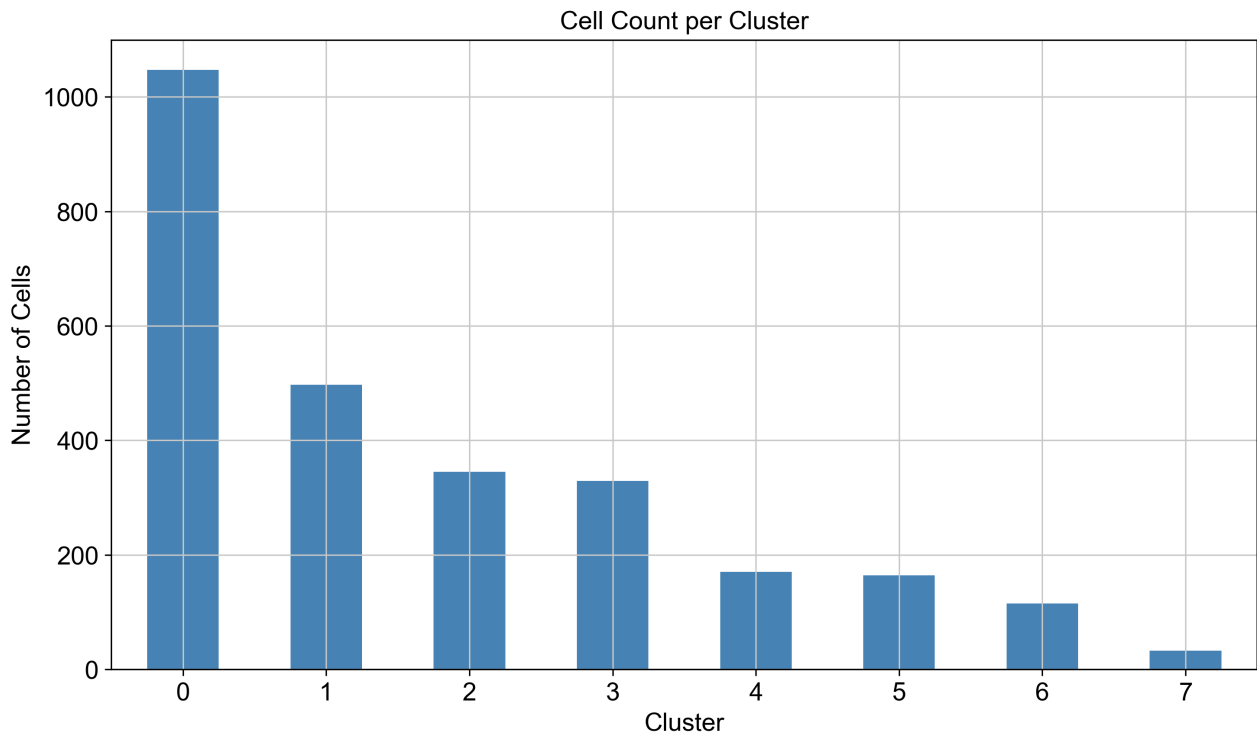
UMAP QC Metrics

UMAP colored by quality control metrics



Cluster Composition

Number of cells per cluster



Cluster Composition

Cluster	Cell Count	Percentage
Cluster 0	1,047	38.8%
Cluster 1	497	18.4%
Cluster 2	345	12.8%
Cluster 3	329	12.2%
Cluster 4	170	6.3%
Cluster 5	164	6.1%
Cluster 6	115	4.3%
Cluster 7	33	1.2%

